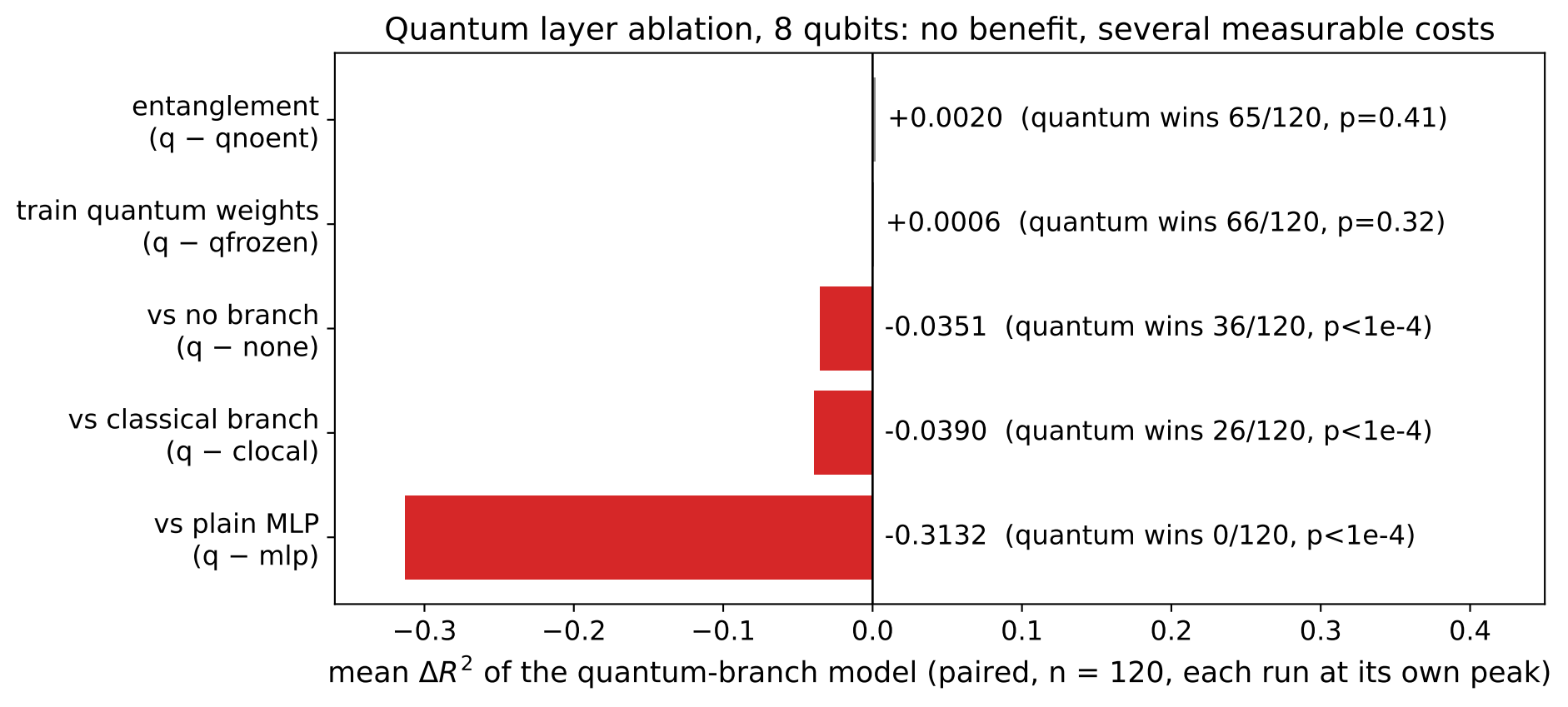
Quantum layer ablation, 8 qubits: no benefit, several measurable costs
entanglement
(q − qnoent)
+0.0020 (quantum wins 65/120, p=0.41)
train quantum weights
(q − qfrozen)
+0.0006 (quantum wins 66/120, p=0.32)
vs no branch
(q − none)
-0.0351 (quantum wins 36/120, p<1e-4)
vs classical branch
(q − clocal)
-0.0390 (quantum wins 26/120, p<1e-4)
vs plain MLP
(q − mlp)
-0.3132 (quantum wins 0/120, p<1e-4)
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
mean ΔR² of the quantum-branch model (paired, n = 120, each run at its own peak)